ANNUAL REVIEWS

Annual Review of Animal Biosciences

Integrated Genomic Approaches to Characterize and Mitigate Heat Stress in Poultry

Carl J. Schmidt[1] and Susan J. Lamont[2]

[1]Department of Animal and Food Sciences, University of Delaware, Newark, Delaware, USA; email: schmidtc@udel.edu

[2]Department of Animal Science, Iowa State University, Ames, Iowa, USA; email: sjlamont@iastate.edu

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

ws.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

Annu. Rev. Anim. Biosci. 2025. 13:121–41

First published as a Review in Advance on October 30, 2024

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-111523-102021

Copyright © 2025 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

OPEN ACCESS

Keywords

heat stress, poultry, genomics, resilience, gene editing

Abstract

With the burgeoning human population, climate change, and expansion of poultry production in hot climates, it is imperative to aid global food security by enhancing the resilience of thermally challenged poultry. As a complement to management approaches used to mitigate heat stress, we give selected examples of recent studies on heat stress in poultry using various omics technologies. An integrated analysis of positional and functional candidate genes is provided, highlighting the most prominent pathways involved in the heat stress response. We finish by discussing efficient strategies to enhance thermal tolerance of poultry by genomics approaches, advocating for preservation of biodiversity that may provide beneficial allelic variation, and identifying current and future challenges in producing climate-resilient poultry.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

INTRODUCTION

During the twenty-first century, climate change is expected to cause worldwide elevated temperatures, with the current UN Climate Change Report (2023) indicating that elevated temperature extremes during heat waves are "virtually certain (99–100% probability)" (1). These extreme temperatures will adversely influence food production, in particular production of animal protein. Worldwide, poultry has become the largest source of animal protein. In 2022, the top 10 countries produced approximately 17 billion chickens, yielding 112 million tons of meat and 1.2 trillion eggs. By 2100, the world population is expected to exceed 10 billion, and providing food security for this population will require increasing poultry production.

One impact of climate change on poultry will be heat stress that will cause production losses due mortality and morbidity. In addition to pathology directly induced by heat, heat stress also causes immunosuppression in both broilers and layers (2), leaving them more susceptible to infectious pathogens. Climate change is expected to shift ranges of pathogens and potentially result in a new era of emerging, reemerging, and endemic pathogens (3).

Heat stress occurs when the internal temperature exceeds the organism's thermoneutral zone—the range of temperature in which the organism does not need to expend extra energy to regulate its body temperature. For chickens, the thermoneutral zone is between 18°C and 24°C (4), and once the temperature exceeds 27°C, the birds need to reduce their heat burden. Many responses that improve the bird's chances of surviving heat stress reduce meat and egg production.

Sensing heat stress in vertebrates begins with the activation of transient receptor potential cation channels in peripheral tissues. These sensory data are transmitted to the parabrachial nucleus of the brain stem via the spinal cord. This nucleus conveys temperature information to the preoptic area of the hypothalamus, which then signals other physiological systems (**Figure 1**). The signals lead to various responses, including reduced feed intake, reduced movement, panting, wing spreading, and increased drinking.

Another response directed from the preoptic area is increased skin vasodilation that diverts blood to the skin and improves heat loss. Vasodilation, along with visceral vasoconstriction, occurs particularly in the intestine. The vasoconstriction is significant, because it can lead to loss of intestinal tight junction integrity, with subsequent leaking of bacterial toxins into the bloodstream (5). This "leaky" gut can then lead to systemic inflammatory responses. The hypothalamus also releases corticotropin-releasing hormone,[1] directing the pituitary to release adrenocorticotropin. Subsequently, the adrenocorticotropin acts on the adrenal glands to promote release of corticosterone, resulting in various responses including decreased fecundity and immune function along with elevated plasma levels of glucose, fatty acids, and amino acids. At the cellular level, heat stress leads to uncoupling of the electron transport chain to reduce heat production, but this results in increased levels of reactive oxygen species (ROS) that can cause cellular damage.

At the molecular level, heat stress leads to protein denaturation and misfolding. This can inhibit many cellular processes, including proliferation, signal transduction, membrane structure, translation, and vesicle transport. In addition, the production of ROS, which can oxidize cellular components including DNA (7), proteins (8), and lipids (9), results in an increased mutation rate, modification of protein function, and altered membrane properties. If damage is sufficient, cell death pathways are activated, leading to apoptosis or necrosis.

The poultry industry uses several approaches to ameliorate heat stress, including environmental management, nutritional, and water interventions, along with feed and water supplements

[1]The format and terminology for chicken genes and gene products are per the International Chicken Gene Nomenclature Committee Report (6).

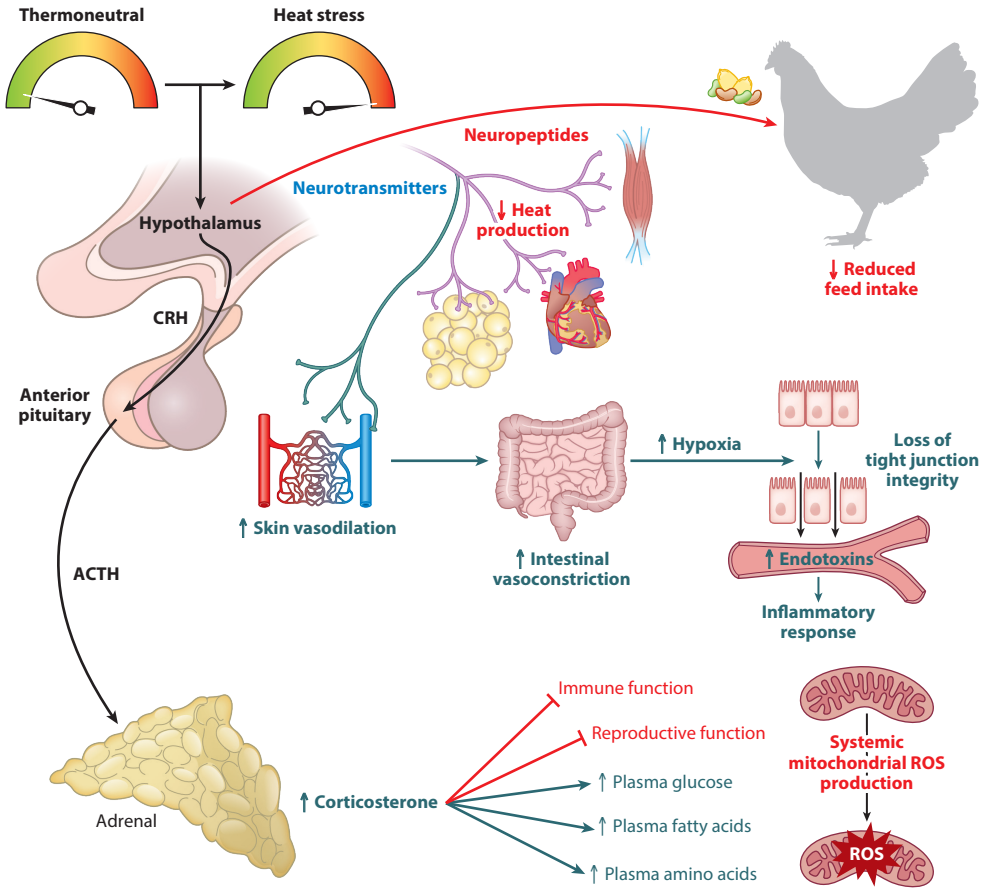

Figure 1

Chicken anatomical and physiological responses to heat stress. Not depicted are behavioral responses, which include panting, lethargy, loss of appetite, wing spreading, and increased water consumption. Abbreviations: ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone; ROS, reactive oxygen species.

(10, 11). Mechanical interventions include ventilation with fans, along with water misters. However, these interventions' effectiveness is limited during periods of high humidity. Nutritional interventions include reducing feed availability during peak heat stress times, thereby reducing metabolic rate and consequent heat production. Other approaches include modulating total protein, free amino acids, carbohydrate, and fat content in diets to lower the bird's metabolic rate (12). Supplements such as vitamin C, vitamin E, selenium, zinc, and chromium can also alleviate the negative effects of heat stress on production traits (13).

Heat stress is categorized as either acute or chronic, and responses differ between the two. Acute heat stress involves a rapid rise in external temperature beyond the thermoneutral range that resolves in a matter of hours. Response to acute heat stress includes a rise in transcripts and proteins encoding the heat shock proteins (HSPs) along with other gene products that determine the organism's outcome to the elevated temperature. Chronic heat stress results from external temperatures remaining above thermoneutral for an extended period, typically days to weeks. The organism's response to chronic heat stress is largely acclimatization, with changes in the transcriptome, proteome, and metabolome relative to the response to acute challenge.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

ORGANIZATION

The following sections group together the type of investigative approaches used to detect associations with heat stress responses in poultry at biochemical, cellular, tissue, and whole-organismal levels. Genome-wide association studies (GWAS) test for statistical associations of structural variants with heat stress–related phenotypes to identify genomic loci (quantitative trait loci, QTL) that control stress responses. Other omics studies characterize how high temperature modulates the bird's response at the level of expression of genes, proteins, or metabolites. Additionally, the impact of differences in gene expression or phenotypes that are regulated by non-DNA variation—epigenetics—is actively researched in poultry heat stress models.

An important aspect of these omics approaches is that they constitute discovery-based science, as contrasted with traditional hypothesis-driven studies. Discovery science is particularly applicable and valuable when generating data at a scale and depth that was previously unattainable. All omics studies generate big data sets. Advances in science and technology enable the integration of the genome, transcriptome, proteome, and metabolome data such that previously undiscoverable effects and responses can be detected (14). Progress of discovery science through to translational use has been pioneered in the human health field (15).

For poorly characterized phenotypic responses, such as heat stress in poultry, the discovery science of large-scale omics enables the unmasking of unanticipated response traits. Unencumbered by traditional thought (hypotheses), discovery science and inductive reasoning can discover novel patterns, associations, and correlations, which can then be used to generate specific hypotheses to test. Integration of discovery research and hypothesis-driven research forms a powerful approach to translate basic discovery into useful application. This is especially valuable for the study of heat stress in poultry, species for which omics tools are advancing rapidly and an environmental stressor that is on the increase.

This article concludes by describing the opportunities to improve thermal tolerance in poultry via genomic approaches. This discussion suggests methods to accelerate genetic gain for heat stress resilience, the crucial need for preservation and study of biodiverse populations, and the challenges likely to be faced. It also recommends consideration of the local climate, environment, and cultural context, both in research studies and in adapting the genotypes of birds to better tolerate stressful temperatures.

GENOMIC STRUCTURAL VARIATION ASSOCIATED WITH THERMAL STRESS RESPONSE

GWAS identify statistically significant associations between genetic variants and phenotypes. They rely on historical recombination events that occurred in the studied population. A related approach, QTL mapping, relies on recombination events that occur during the experiment, for example, using an F_2 cross of phenotypically and genetically diverse lines. Discovery-driven identification of structural variants that are statistically associated with heat stress phenotypes can be translated into application in multiple ways, including

- genomic selection for variants beneficially associated with heat stress response by commercial breeding companies,
- positional candidate gene identification for subsequent hypothesis-driven functional studies by researchers, and
- characterization of molecular biodiversity for genetic management by conservationists and hobby breeders.

Adaptation to environmental changes is a polygenic trait. Quantitative genetics focuses on phenotypes and considers adaptation to occur via incremental allelic frequency changes, whereas

population genetics assumes phenotypic adaptation via single-locus mutations (16), detectable as selective sweep regions of reduced genetic heterogeneity. Defining the architecture of adaptation traits in experimental populations facilitates expansion to the study of the complexity of natural populations. Many studies limit the number of adaptive alleles by initiating experimental populations with few founders to aid detection of the adaptive alleles and subsequent hypothesis-driven functional studies (16). A shortfall of an early generation cross such as an F_2 is the small number of meiotic recombinations that have occurred and thus the large blocks of linkage disequilibrium in which causal mutations and marker loci colocate. This makes it difficult to narrow down which of many genes near statistically significant markers may truly control the phenotypes of interest.

To address the shortfall in mapping resolution of early crosses, heat stress studies were conducted by using the F_{18} and F_{19} generations of a highly divergent cross, which was initiated with one broiler rooster mated to 6 highly inbred Fayoumi hens. Broilers are relatively heat susceptible and fast growing and have large body size, whereas Fayoumi (originating in Egypt) are heat resilient and slow growing and have small body size. Using a 600K single-nucleotide polymorphism (SNP) panel, Van Goor et al. (17) identified many QTL regions for blood components, which were lowly to moderately heritable, and functional analysis of genes in these regions indicated the angiopoietin pathway as likely involved in response to heat stress. The angiopoietins are vascular growth factors for both embryonic and postnatal angiogenesis. Activation of this pathway in heat stress may represent a mechanism to increase vascularization and thereby thermoregulation through additional blood vasculature. Analysis of body weight, breast meat yield, and feed digestibility showed that these phenotypes were all moderately heritable, and QTL regions were identified for each trait. A single window on chromosome 1 explained more than 15% of the genetic variation for breast meat yield under heat stress conditions, suggesting that a major gene for muscle deposition in hot climates exists in this region (18).

Selection signatures or selective sweeps in molecular population genetics are the genomic areas of reduced DNA polymorphism around the locus of a recently fixed beneficial allele. Because of linkage disequilibrium, the neutral-effect alleles near the causative mutation exhibit a hitchhiker effect and show a reduction in heterozygosity similar to that of the fixed allele. Detection of signatures of selection in indigenous populations can elucidate genetic variants that reflect adaptation to the environment in which the population has been subject to long-term natural selection. Thus, selective sweeps in populations from hot climates may reveal genomic regions in which beneficial alleles are located for heat resilience.

Selection sweep analysis of five breeds or ecotypes of chicken indigenous to Africa were conducted using a 600K SNP panel. Egyptian chickens had selection sweeps encompassing genes for adaptation to heat, solar radiation, ion transport, and immunity. East African populations that are adapted to high altitude showed a selection signature with genes for angiogenesis and oxygen–heme binding and transport (19). More than 700 chickens from various populations in Africa and Northern Europe were evaluated with a 600K SNP panel to determine genomic regions that might differentiate populations with a natural selection history in hot versus cool climates, respectively. The selective sweeps showed stronger selection for stress signaling and angiogenesis in African chickens and stronger selection for energy homeostasis in European chickens (20). Selective sweep regions identified by genomic resequencing of native Chinese chickens in hot, arid, and harsh environments also identified regions related to the circulatory system and blood vessel development, as well as apoptosis, central nervous system development, fatty acid metabolism, and response to stimuli (21). Whole-genome resequencing of samples of chicken populations from varied tropical climates showed diverse genomic locations of selective sweeps, suggesting that environmental adaptation to hot climates can be accomplished differently in different populations, using unique genes and pathways to achieve the same level of fitness in their environments (22).

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

TRANSCRIPTOME ANALYSIS

The cellular response to heat stress is triggered by (*a*) misfolding or denaturation of proteins along with (*b*) changes in the membrane fluidity altering signaling pathways. These events lead to changes in transcription, mRNA stability, translation, protein stability, epigenetic markings, and metabolism. Transcriptome analysis can identify and quantify individual mRNAs and noncoding RNAs in a given sample. Differential gene expression (DGE) analysis, a hypothesis-free method, can then be used to identify genes that are either positively or negatively responsive to different stimuli.

DGE analysis has defined genes responsive to heat stress in poultry and has shown that different cells and tissues can have distinct responses to thermal challenge. The identity and number of responsive genes vary between tissues. In addition, gene expression levels may be upregulated in one tissue and downregulated in another in response to heat stress. Also, the duration of heat stress determines the expression level of some genes with acute (<1 day) or chronic (>1 day) heat stress, yielding distinct responses (23). For example, liver expression of both HSP90AA1 and HSPA2 is significantly elevated following 4 h of heat stress. However, after 6 weeks of chronic heat stress, the mRNA levels of both genes are similar to those seen in livers from control birds. In contrast, mRNA levels of HSP90AA1 and HSPA2 were elevated in muscle under acute and chronic stress. In the heart, HSPA2 was upregulated under both conditions, whereas HSP90AA1 levels under chronic heat stress were comparable to control hearts.

DIFFERENTIALLY EXPRESSED GENES

To identify common processes regulated by heat stress, a list of heat-responsive genes was assembled from published work that identified DEGs in heat-challenged cell lines or tissues isolated from cattle (24–26), pigs (27–29), chickens (30–46), ducks (47–49), or turkeys (50, 51). This list was used to examine Gene Ontology (52) terms and identify common biological processes responsive to heat stress across these livestock species (**Figure 2**). The major group of responsive genes encode chaperones, which are found in the protein folding, response to unfolded protein, response to abiotic stimulus, response to endoplasmic reticulum stress, and response to environmental stimulus ontology groups. Chaperones are found throughout the cell, spanning from the plasma membrane to nuclear compartments. Chaperones assist their client proteins in achieving or maintaining their correct 3D structure (53). They also assist in refolding improperly folded or denatured proteins and facilitate degradation of proteins via either the proteasomal or autophagy pathways. Typically, chaperones are expressed and function at a basal level under thermoneutral conditions. Heat stress increases transcription and translation of this class of proteins to promote protein homeostasis during thermal challenge.

The first identified chaperones were named HSPs because their levels increased in response to heat stress. These proteins arose from evolutionarily conserved families of related genes that are classified by the size of the encoded proteins (53), including HSP100, HSP90, HSP70, HSP60, DNAJ81 (HSP40), HSP110, and small HSPs (ranging in size from 10 to 40 kDa). These families interact to form a complex network, with members of the HSP70 (70 kDa) and HSP90 (90 kDa) families serving as hubs (54). In addition to the HSP families, numerous other proteins serve similar functions, sometimes with specific clients. For example, a family of tubulin-folding cofactors (TBCA, TBCC, TBCD, TBCE) assist beta tubulins in arriving at their correct 3D structure (55).

HSPs also control the activity of heat shock factors (HSFs). HSFs are a family of transcription regulators that activate heat stress–responsive genes. Members of the HSP90 and HSP70 families operate in a regulatory loop, binding and inactivating HSFs under normal conditions. During heat and other types of stress, the HSPs are recruited to act as chaperones for client proteins.

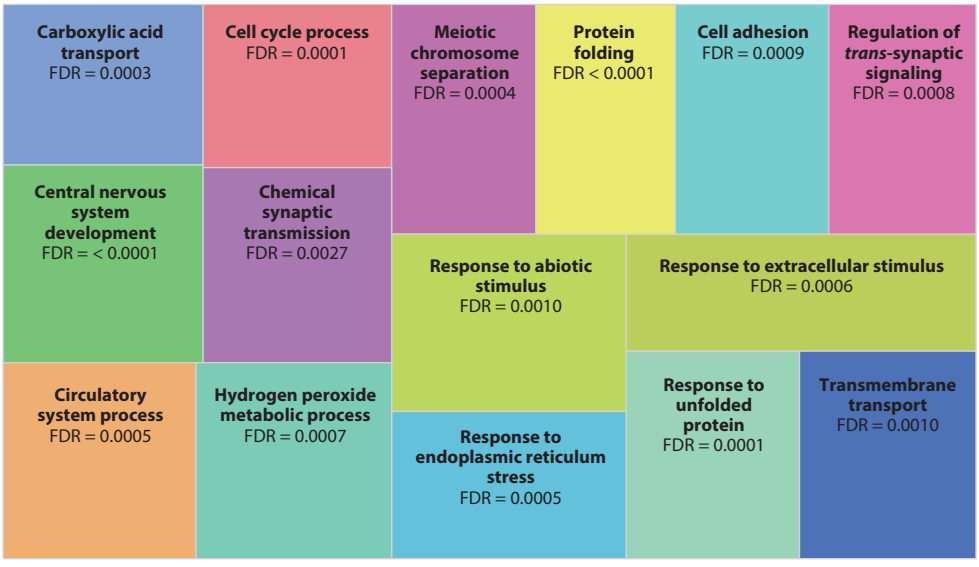

Figure 2

Gene Ontology biological process terms and false-discovery rate (FDR) for heat stress–responsive genes identified in two or more species.

Consequently, HSFs are released, enter the nucleus, and activate heat-responsive promoters. As conditions return to normal, HSPs are no longer needed to chaperone their clients and are freed to bind and inactivate HSFs. This feedback loop tightly couples environmental conditions to the heat stress response.

Heat stress profoundly affects the cell cycle biological process. Heat-responsive genes impact multiple steps in the cell cycle, including cell cycle checkpoints, DNA synthesis, DNA repair, and cytokinesis. Generally, genes that inhibit progression through the cell cycle are upregulated by heat, whereas those that promote progression are downregulated. During heat stress, the electron transport chain is uncoupled to reduce endogenous heat production, but a negative consequence is the production of oxygen radicals that damage DNA, proteins, and membranes. Cell cycle inhibition during heat stress provides time for DNA repair mechanisms to act, chaperones to assist in refolding proteins, and enzymes to repair membranes prior to cytokinesis. In addition, another group of transcripts enriched during heat stress encode enzymes that remove hydrogen peroxide. These enzymes function to limit the cellular damage caused by oxidative stress.

Heat-responsive genes are also associated with the Gene Ontology terms central nervous system development, regulation of *trans*-synaptic signaling, and chemical synaptic transmission. Most of these genes are associated with synaptic signaling of axons and dendrites. Of the 60 genes that are annotated with the ontology term synapse, 29 impact glutamate receptor function. Glutamate receptors play an important role in visceral sympathetic regulation of the heat stress response by conveying thermal information from the skin to the hypothalamus (56). This leads to increased heart rate and elevated cutaneous blood flow that facilitate heat loss while inhibiting thermogenesis in adipose tissue (56).

The carboxylic acid transport and transmembrane transport ontology groups are predominated by solute carrier (SLC) amino acid transporters. Two classes of upregulated transcripts that impact the heat stress response include arginine (SLC7A1 and SLC7A7) and glutamine (SLC38A3) transporters. Supplementing feed with arginine during heat stress improves production yields in

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

poultry (57, 58). This may result from the importance of arginine to nitric oxide production and from improving intestinal barrier function (59). Glutamine has also been shown to improve intestinal barrier function, elevate HSP70 gene products, and improve production traits during heat stress (60–63).

A major process of cell adhesion that heat stress impacts is likely cell motility. DEGs regulated by heat stress include WNT family member 4 (WNT4), secreted frizzled-related protein (SFRP1), thrombospondin 1, and SMAD associated family member 3. WNT4 controls multiple biological processes, including promoting cell motility (64, 65), whereas SFRP1 negatively regulates WNT function. This combination of positive and negative regulators of WNT activity suggests that regulation of cell motility during heat stress is carefully balanced. Thrombospondin (66, 67) can either promote or inhibit motility based on cell type, whereas SMAD associated family member 3 couples tumor growth factor B (TGFB) activity to increased cellular motility (39). Enrichment for cell motility is consistent with reports that HSF1 and members of the HSP70 family also promote cell migration (68, 69).

Genes responsive to heat stress that affect circulatory system processes are largely involved in heart development along with angiogenesis. With respect to heart development, the biological processes identified include chamber morphogenesis, cardiac septum morphogenesis, and cardiomyocyte differentiation. Terms associated with angiogenesis affected by heat stress include artery development, sprouting angiogenesis, and tube development. Included in this group are genes that function in the TGFB and BMP pathways, which converge on the same downstream effector proteins. Because the ligands, receptors, and effectors can vary between cell types, modulation of the TGFB and BMP pathways by heat is likely to cause distinct responses in different cells.

INTEGRATING GENOME-WIDE ASSOCIATION STUDIES AND CANDIDATE GENES FROM EXPRESSION STUDIES

GWAS associate traits, such as response to heat stress, to specific regions of the genome. Traits that have been mapped include heat tolerance, as measured by comparing birds that survived or died due to accidental heat stress (70, 71). Controlled heat stress studies have identified body temperature, body weight, breast muscle yield, digestibility, egg quality, and feed efficiency as responsive to thermal challenge (18, 41, 72, 73). GWAS of blood components using the i-STAT blood analyzer (17, 74, 75) have identified quantitative traits in response to heat stress that include hematocrit, partial pressure of oxygen, total carbon dioxide, bicarbonate levels, glucose levels, pH, potassium levels, and base excess (17, 74, 76).

The GWAS identify regions of the genome associated with specific traits, and genes within these regions are positional candidates for affecting their respective traits. The candidate list can be narrowed further by identifying those genes that exhibit differential expression due to heat stress (77). Categorizing the candidate genes to Gene Ontology terms helps define the biological processes affecting the mapped traits (**Figure 3**). Most of these genes are involved in more than one process, which can be summarized as acting as chaperones, regulators of immune responses, and controlling glial cells. The broad impact of these candidates is consistent with the broad impact heat stress exerts on an organism. The most pleiotropic genes are HSP90AA1 (affects 10 of 13 processes), followed by Toll-like receptor 4 (TLR4) and HSPA4 (both affect 9 of 13 processes). In combination, differential transcriptional regulation of these three genes could have an expansive and complex impact on the heat stress response.

Several of the candidate genes point to significant negative effects of heat stress. Due to uncoupling of the electron transport chain, heat stress results in the production of superoxide radicals. Superoxide dismutase 1 is responsible for removing these radicals, thereby preventing damage to

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

Figure 3

GO biological process terms, gene symbols, and false-discovery rate for heat-responsive genes that overlap with candidate genes identified by genome-wide association studies. GO terms were prioritized and uniqueness defined by REVIGO (118). Abbreviations: FDR, false-discovery rate; GO, Gene Ontology; IFNB, interferon B; MYD88-ITLRSP, MyD88-independent-Toll-like receptor signaling process.

cellular components. The TLR4 gene product is part of the innate immune system responsible for activating inflammatory pathways in response to the bacteria endotoxin lipopolysaccharide (LPS). One of the important negative effects of heat stress is leaky gut, which is caused by weakening of the intestinal tight junctions. LPS is normally restricted to the lumen of the gut, but increased intestinal permeability allows LPS to spread systemically, activating TLR4 and initiating an inflammatory response. If unchecked, this inflammatory response can ultimately cause mortality. SLC7A2 is an amino acid transporter responsible for importing arginine, an important precursor to nitric oxide. Nitric oxide can be both beneficial and harmful. It promotes vasodilation, thereby promoting heat loss, but at the same time it can promote oxidative damage to the cell.

METABOLOMICS

Metabolomic approaches use either mass spectroscopy or nuclear magnetic resonance to identify metabolites in biological samples. Several studies focusing on either liver or breast muscle have reported the metabolic response to heat stress in poultry (2, 30, 78, 79). Several metabolites have been identified in common between these studies.

One of the sequelae to heat stress is oxidative stress. Thus, unsurprisingly, a common response is changes in pathways that offset the appearance of oxygen radicals. Changes in the concentration of metabolites involved in glutathione production are typically observed in these studies and

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

include responses in the levels of cysteine, glutamylcysteine, reduced glutathione, and oxidized glutathione (30, 32, 63, 80). Maintaining the reducing power of glutathione requires additional levels of NADPH (nicotinamide adenine dinucleotide phosphate). Heat stress elevates levels of 6-phosphogluconate, a metabolite of the pentose phosphate pathway. 6-Phosphogluconate is a substrate for phosphogluconate dehydrogenase, and during catalytic production of the product ribulose-5-phosphate, phosphogluconate dehydrogenase converts NADP+ to NADPH. Other metabolites responding to heat stress that offset oxidative damage to liver and skeletal muscle include carnosine, anserine, and betaine. Taken together, the response of these metabolites points to the importance of offsetting oxidative stress to maintain homeostasis in the face of heat stress.

Heat stress also significantly affects gluconeogenesis. This is emphasized by increased levels of glucose-6-phosphate, fructose-6-phosphate, and glucose in the livers of birds subjected to thermal challenge. This is consistent with the need to provide glucose to other tissues due to the increased energy needed to maintain homeostasis (81). Furthermore, liver lipid metabolism is altered with elevated levels of myristate, myristoleate, palmitoleate, and glycerol-3-phosphate. These are essential components of triacylglycerol production, indicating that the liver likely provides this energy source to other tissues during heat stress.

These metabolic data show the importance of such studies to understand the heat stress response. In particular, they indicate specific components of pathways that are altered as elevated temperatures challenge homeostasis. In poultry, these studies have focused on liver, skeletal muscle, and plasma responses. Expanding similar metabolic analysis to other tissues will be important to providing a systems-level understanding of the heat stress response.

PROTEOMICS

Proteomics approaches have also been used to identify proteins that are differentially regulated by heat stress. Although it yields lower numbers of identified molecules as compared with metabolomics or transcriptomics, the approach both supports and extends our understanding of the heat stress response. Proteomic analyses have been conducted using heat-stressed chicken liver (63, 82), hypothalamus (80, 83), adrenal gland (84), spleen (85), and testes (5), along with duck skeletal muscle (86) and liver (87). Proteomic analysis has identified multiple chaperones as differentially expressed proteins during heat stress. Among these are HSP25, HSP90AA1, HSPA5, HSPA8, HSPA9, HSPB7, HSPD1, and HSPE1 (5, 84). Non-HSP chaperones identified as differentially expressed proteins include TRAP1, SERPINH1, CALR, PDIA1, PHB1, and MYBPC (88).

In addition to chaperones, levels of several metabolic enzymes affecting pathways are altered by heat stress. For example, heat stress typically upregulates levels of peroxiredoxin-1, superoxide dismutase 1, glutathione *S*-transferase, and glutathione peroxidase in the liver (63, 89). These enzymes remove oxygen radicals from the cell, thereby preventing oxidative damage. Heat stress also impacts carbonic anhydrase (CA5 and CA6) levels (86, 89). Birds pant as their main mechanism for cooling, which makes them particularly susceptible to respiratory alkalosis during heat waves. CA is responsible for maintaining blood pH levels by hydrating carbon dioxide to bicarbonate ions and protons. Consequently, CA levels are likely responding to heat stress to maintain homeostatic pH levels.

Isocitrate dehydrogenase (IDH), a critical enzyme of the tricarboxylic acid (TCA) cycle, is downregulated (63). This would reduce flux of metabolites through the cycle, limiting electron delivery to the oxidative phosphorylation pathway and reducing heat output. In addition, lowered IDH activity reduces α-ketoglutarate levels. This metabolite not only functions in the TCA cycle but regulates the activities of the ten-eleven translocation proteins, the activity of which is critical

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

to epigenetics by functioning in the DNA demethylation pathway. Hence, IDH couples an important metabolic cycle to epigenetic regulation. Another downstream effect is that a decrease in α-ketoglutarate leads to lower levels of succinate. Overall reduced TCA activity also reduces the level of damaging ROS produced by the electron transport chain. Consequently, reducing succinate levels lowers the amount of ROS generated by the electron transport chain (90).

Another heat-induced differentially expressed protein is TLR4 (91). Endotoxins such as bacterial LPS released into tissues due to loss of tight junction integrity during heat challenge will activate the TLR4 receptor. TLR4 then stimulates the NFKB pathway, resulting in increased levels of interleukin 1B, interleukin 6, and tumor necrosis factor A mRNAs. Proteomics data have verified the increased levels of these and other proinflammatory cytokines in serum during heat stress (85). During prolonged or excessive heat stress, these cytokines can drive the organism to heat stroke and death.

EPIGENETICS

Epigenetics is the scientific field that studies the changes in gene expression that are mitotically and/or meiotically heritable and do not create a change in DNA sequence (92). In recent decades, epigenetic modification of gene expression has been shown to impact long-term phenotypes in livestock, including birds (54, 93).

Thermal conditioning of chick embryos or neonates by exposing them to a period of higher-than-normal ambient temperature has long been explored as a management strategy to improve heat resilience of birds later in life. It is now accepted that the mechanism of action resulting in this improved phenotype is epigenetic modification of gene expression. Epigenetic thermal conditioning assumes that environmental factors such as ambient temperature greatly influence the set point for homeostatic mechanisms during critical developmental phases before or soon after hatching.

Most studies have been conducted on broilers because of their short time to market age and their rapid growth rate, which makes them particularly susceptible to heat stress. In a study aimed at optimizing conditions of post-hatch thermal treatment, day 3 post hatch and temperatures of 36.0–37.5°C optimized performance traits, including higher body weight, lower feed intake, better feed efficiency, lower heat-related mortality, and lower triiodothyronine (94). Post-hatch exposure to a high–temperature humidity index environment reduced heat production of birds when exposed to thermal challenge in later life (broiler market age) by increasing heat loss through convection and radiation, reducing stress levels, and reducing HSPs in lung and heart muscle during thermal challenge (95). Thus, the epigenetic changes induced by thermal conditioning are broadly protective of preserving, or even enhancing, broiler productivity traits during a heat challenge.

Given the development of the embryo in the egg outside of the hen's body, the chicken is an ideal model for understanding epigenetic mechanisms (92). The ability to apply uniform thermal treatments to thousands of embryos in eggs simultaneously in artificial incubation also makes embryonic thermal conditioning an attractive, practical option to apply in the poultry industry to improve bird welfare and productivity.

The mid-incubation period is crucial for development of important neuro-endocrine circuits in the embryo, including thermoregulation (hypothalamus–pituitary–thyroid axis) and the stress response (hypothalamus–pituitary–adrenal axis) (96). To date, the two most-studied major epigenetic marks associated with adaptation to heat stress in chickens are DNA methylation and histone posttranslational modifications. The alterations in cellular properties of the frontal hypothalamus during development, as needed to achieve plasticity, are achieved by silencing or activating gene transcription via regulation by histone modifications. A thermal conditioning study in chickens

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

demonstrated induction of global histone H3 lysine 27 (H3K27) demethylase in the hypothalamus and also demethylation of the promoter of the brain-derived neurotrophic factor gene. Knockdown of the H3K27-specific methyltransferase (EZH2) transcript inhibited brain-derived neurotrophic factor mRNA expression and disrupted the set point establishment for thermoregulation. The study's findings highlight the specific epigenetic role of chromatin modifications in establishment of heat resilience (93).

In a study of embryonic thermal conditioning that showed improved heat tolerance of broilers in a later heat challenge, the *pectoralis major* muscle, the largest and most economically valuable muscle, showed DGE between control and thermally conditioned birds. In comparison to controls, the thermally conditioned birds had more differentially expressed genes involved in vascularization and muscle growth, cell proliferation, energy metabolism, and mitochondrial function. They also expressed more pathways, including those related to vascularization, stress response, antiapoptotic and epigenetic processes, and metabolism (97). A subsequent study investigated the hypothalamus and muscle tissue for genome-wide distribution of two histone posttranslational modifications, H3K4me3 and H3K27me3, which have been reported to contribute to environmental adaptation. The histone differential peaks in the hypothalamus included genes involved in neurodevelopmental, gene regulation, and metabolic functions. However, few differences were found in the muscle tissue. These results demonstrate that the response to embryonic thermal manipulation is most likely mediated by epigenetic changes in the hypothalamus, which may have distal effects in the body and contribute to thermal resilience in later life (98).

Many fewer studies have been conducted on thermal conditioning and epigenetics in layer chickens than in broilers. Layer chicks were exposed to 37°C for one day at day 5 of age (thermal conditioning), or injected with LPS, and then challenged at maturity with the same environmental stressor. Thermal conditioning did not affect the hens' antibody response, endocrine parameters, or layer performance. In contrast, early-life exposure to LPS altered the level and kinetics of antibody binding to both LPS and keyhole limpet hemocyanin, an antigen to which the birds had not been exposed previously. Because heat stress is often accompanied by leaky gut, in which LPS enters the body from the gut lumen, early exposure to LPS may protect against one of the detrimental outcomes of heat stress (99).

IMPROVING THERMAL TOLERANCE

In the face of rising temperatures, it will be important to increase the thermal tolerance of poultry. By examining existing lines of birds, allelic variants can be identified that improve their ability to withstand heat stress. Two classic variants with improved heat tolerance are Naked Neck and Frizzled (100). The Naked Neck locus eliminates feathers from the neck, resulting in improved heat loss. Frizzled has a similar effect by reducing feather density. Although these variants are common in backyard flocks in tropical regions, their negative impact on production traits makes them incompatible with the poultry industry.

There is genetic evidence that many of the responses to heat stress have sufficient heritability that selection could improve birds' ability to tolerate this challenge (70, 72, 101, 102). For example, direct selection for improved egg traits over two generations under heat stress successfully increased egg production and eggshell strength. Concurrent with these improved traits was increased expression of the HSP90 mRNA, suggesting that increased expression of HSP genes may improve heat tolerance.

Allelic variations in HSP genes are also associated with thermotolerance. For example, by comparing between heat-tolerant Lingshan and intolerant White Recessive Rock breeds, Chen et al. (103) identified a SNP in the 5′ flanking region of the HSP90B1 gene associated with

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

heat tolerance. The allele conferring tolerance was associated with elevated levels of HSP90B1 transcript. In a similar study, allelic variants in the 5′ flanking region of the HSPA5 gene were associated with elevated HSPA5 mRNA levels and improved heat tolerance (104). Another means to improve heat stress response could be to increase the copy number of HSP genes. For example, thermotolerance increased linearly with HSP70 gene copy number in *Drosophila* (105). Taken together, the data suggest that increasing the basal level of HSP90 or HSP70 expression could improve the thermotolerance of poultry.

PRIMORDIAL GERM CELLS, GENOME EDITING, AND BIOBANKING

Climate change is likely to increase the poultry industry's need for heat-tolerant birds. In theory, this could be accomplished by introgression of desirable alleles from thermotolerant birds. In reality, the industry is unlikely to adopt this approach, because introgression will bring along hitchhiking alleles that reduce the quality of other important production traits. A far more efficient means to introduce new alleles would be to use gene editing technologies to modify specific sites in the genome. Although gene editing would facilitate rapid integration of beneficial genetic variants, consumer hesitancy or lengthy approval processes to satisfy governmental regulations may slow the delivery of the benefits of these technologies to the public.

The most common gene editing approach used currently is clustered regularly interspaced palindromic repeats (CRISPR)-Cas9 technology. CRISPR-Cas9 can introduce a variety of mutations, including deletions, substitutions, insertions, and complex rearrangements in the genome (106). Additional CRISPR enzymes have been isolated or genetically engineered to allow for insertion of large DNA fragments encompassing entire genes (107) along with numerous other applications (108).

Currently, this approach requires establishing primordial germ cells (PGCs) from the chicken lines intended for editing (at this time, the approach has been successfully applied only to the chicken). Once established, the genome of the PGC can be edited to introduce the desired allele(s) and microinjected into embryos of a surrogate host (109). If successful, the PGCs will migrate to and populate the gonads of the recipient and produce viable gametes. Once they become adults, these gene-edited birds can then be crossed to generate the desired line of birds. To facilitate this process, a line of chickens has been developed containing an inducible Caspase 9 gene specifically expressed in germ cells. When induced, Caspase 9 triggers apoptosis in all of the germ cells (110). This Caspase 9 line is an excellent surrogate for receiving PGCs, as the microinjected cells will not have to compete with any surrogate host germ cells.

The poultry industry uses multiple elite lines that ultimately give rise through crossbreeding to commercial meat or egg-laying birds. The birds of the elite lines may be great-great-grandparents of the birds that go into the production houses. The first step to using gene editing would be to establish PGC lines from these elite birds. These PGCs would provide the stock for targeted gene editing to improve the tolerance of the chickens. Initial editing might focus on introducing the alleles that have been identified in heat-tolerant birds (see above) or increasing the copy number of specific HSP genes. A reasonable hypothesis is that if the edit confers thermotolerance, the edited PGCs should be heat resistant compared with the parental PGCs. This hypothesis suggests that a large library of distinct gene edits could be tested in vitro to identify those PGCs that are thermotolerant. Only those PGCs carrying edits that pass this test would be used to create gene-edited chickens. This would significantly reduce the cost and time invested to obtain thermotolerant elite birds that could go back into the breeding program.

Not only does the ability to culture chicken PGCs in vitro open the door to gene editing, the cultured PGCs provide an important means to preserve genetic diversity. Cryopreserved PGCs

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

can be thawed, allowed to proliferate in vitro, then transferred to a surrogate host (111). This procedure can be used to reestablish lines from the stored cells. This has tremendous value to conservation, as stored PGCs allow maintaining the genetic diversity of the species without the need to maintain flocks of birds. This will be particularly critical to maintaining the allelic diversity found in feral, indigenous, and backyard flocks.

CHALLENGES IN IMPROVING RESILIENCE TO HEAT STRESS IN POULTRY

Although studies have revealed many promising genes and biological pathways involved in response to heat stress in poultry, important challenges exist in the implementation of genetic improvement. Most of the major commercial poultry breeding companies are located in temperate climates. Thus, the environments in which many of their evaluation and selection programs are conducted do not encompass the vast range of climates in which the commercial birds must perform. To address this issue, companies place birds that are related to the breeding candidates into challenging environments around the world to evaluate important genotype-by-environment interactions (112). Breeding objectives may need to be adjusted to include more incorporation of heat-resilience traits into highly productive stocks and greater productivity into locally adapted stocks (113).

In some contexts, management strategies for alleviation of heat stress may be easier to apply than genetic improvement programs. Many options for management of heat stress exist, including feed manipulations (timing; moisture content; altering protein and energy content; adding electrolytes, vitamins, probiotics, and prebiotics) (114). Ideally, genetic improvement can be combined with good management practices.

Poultry are produced in a wide variety of management settings, ranging from facilities with millions of birds in environmentally controlled buildings in a single site in high-income countries to birds raised in semi-scavenging settings with minimal provision of feed and shelter in low-income countries. Although the productivity of birds in the latter management setting may be very low, these poultry provide families with dietary protein and essential micronutrients, and potentially a modest income that helps to empower women and youth (115). Local ecotypes may also have special cultural meaning associated with their specific phenotypes and their use in social events and ceremonies.

Importantly, indigenous populations of poultry are well-adapted through natural selection to their environment, including resilience to heat stress. They thus represent a rich reservoir of genetic biodiversity that can be of value in improving genetic resilience to heat stress if the beneficial variants are identified. This indicates a critical need to maintain global biological diversity to face current and future challenges generated by climate change (115). To accomplish this effectively, the indigenous populations need to be well-characterized, programs need to be in place to assess threats to genetic diversity, and effective mechanisms for in situ and ex situ conservation are required. This includes biobanking to safeguard against loss of endangered biodiversity.

Additionally, comprehensive studies should gather information not only on the poultry themselves but also on their local production environments, including climate parameters, demography, and breeding conditions. These data can be difficult to collect and analyze. One method of integrating these data sets, which are diverse in type and scale, is via use of Geographic Information Systems (116). With rich data sets thoroughly describing local agro-ecologies, environmental niche modeling can identify the main environmental influences that are drivers of selection, including such factors as temperature, precipitation, and soil/land cover (117). Studies combining information on both the genetics of local poultry and the environments

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

in which they are located can, therefore, aid in identifying key components of environmental adaptation.

SUMMARY POINTS

1. Increasing human population, climate change, and expansion of poultry production in hot climates require new ways to enhance heat resilience in poultry to ensure global food security.
2. Concordance of genes identified as positional candidates from genome-wide association studies and functional candidates revealed from transcriptomics studies give confidence in their role in response to heat stress.
3. Scientific discovery research leverages the power of omics technologies, which generate massive data sets, to find biological patterns (i.e., genes, pathways) to further interrogate in hypothesis-driven research.
4. Diversion of blood away from the intestine by heat stress causes weakening of tight junctions and release of lipopolysaccharide to the circulatory system. This drives systemic inflammation that contributes to heat stress morbidity and mortality.
5. Additional responses to heat stress include angiogenesis, oxidative stress, and suppression of immune function.
6. Genome editing of elite lines would be the most efficient way to rapidly introduce beneficial alleles into commercial stocks.

FUTURE ISSUES

1. Allelic variants that support heat resilience may be prevalent in locally adapted breeds but may be absent or in low frequency in current commercial lines of poultry.
2. Populations that are well-adapted to their local environments may be inefficient in terms of meat and egg production efficiency.
3. Geographic and climatic information about the sites where genetic samples were collected should be included in analyses.
4. Putative causative variants should be identified from genome-wide association studies to provide candidates for functional analysis and to improve relevant traits by genome editing.
5. Integration of the various omics technologies is needed to provide systems-level understanding of heat stress and thermal tolerance.
6. Individual cells within a tissue likely respond differently to heat stress, and this would play an important role in survival. Single-cell sequencing needs to be applied to unravel this complex response to heat stress.
7. A fuller understanding of the role of epigenetics in heat resilience is required to apply promising management strategies most effectively, for example, thermal conditioning of embryos and neonates.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

This project was supported by the Agriculture and Food Research Initiative Competitive Grant 2011-67003-30228 from the USDA National Institute of Food and Agriculture. Activities at Iowa State University were also supported by the College of Agriculture and Life Sciences, State of Iowa, and Hatch funding.

LITERATURE CITED

1. Intergov. Panel Clim. Change (IPCC). 2023. *Climate change 2023: synthesis report*. Rep., IPCC, Geneva. **https://digitallibrary.un.org/record/4008074?v=pdf**
2. Huang MY, An YC, Zhang SY, Qiu SJ, Yang YY, Liu WC. 2024. Metabolomic analysis reveals biogenic selenium nanoparticles improve the meat quality of thigh muscle in heat-stressed broilers is related to the regulation of ferroptosis pathway. *Poult. Sci.* 103:103554
3. Baker RE, Mahmud AS, Miller IF, Rajeev M, Rasambainarivo F, et al. 2022. Infectious disease in an era of global change. *Nat. Rev. Microbiol.* 20:193–205
4. Pontara Vilas Boas Ribeiro B, Yanagi T Jr., Duarte de Oliveira D, Ribeiro de Lima R, Gilberto Zangeronimo M. 2020. Thermoneutral zone for laying hens based on environmental conditions, enthalpy and thermal comfort indexes. *J. Therm. Biol.* 93:102678
5. Wang SH, Cheng CY, Chen CJ, Chan HL, Chen HH, et al. 2019. Acute heat stress changes protein expression in the testes of a broiler-type strain of Taiwan country chickens. *Anim. Biotechnol.* 30:129–45
6. Burt DW, Carre W, Fell M, Law AS, Antin PB, et al. 2009. The Chicken Gene Nomenclature Committee report. *BMC Genom*. 10(Suppl. 2):S5
7. Salmon TB, Evert BA, Song B, Doetsch PW. 2004. Biological consequences of oxidative stress-induced DNA damage in *Saccharomyces cerevisiae*. *Nucleic Acids Res.* 32:3712–23
8. Kikusato M, Nakamura K, Mikami Y, Mujahid A, Toyomizu M. 2016. The suppressive effect of dietary coenzyme Q_{10} on mitochondrial reactive oxygen species production and oxidative stress in chickens exposed to heat stress. *Anim. Sci. J.* 87:1244–51
9. Bai YX, Fang F, Jiang JL, Xu F. 2017. Extrinsic calcitonin gene-related peptide inhibits hyperoxia-induced alveolar epithelial type II cells apoptosis, oxidative stress, and reactive oxygen species (ROS) production by enhancing Notch 1 and homocysteine-induced endoplasmic reticulum protein (HERP) expression. *Med. Sci. Monit.* 23:5774–82
10. Daghir NJ. 2009. Nutritional strategies to reduce heat stress in broilers and broiler breeders. *Lohmann Information*, Jan. **https://lohmann-breeders.com/lohmanninfo/nutritional-strategies-to-reduce-heat-stress-in-broilers-and-broiler-breeders/#**
11. Qaid MM, Al-Garadi MA. 2021. Protein and amino acid metabolism in poultry during and after heat stress: a review. *Animals* 11:1167
12. Leeson S. 1986. Nutritional considerations of poultry during heat stress. *World's Poult. Sci. J.* 42:69–81
13. Kim DY, Kim JH, Choi WJ, Han GP, Kil DY. 2021. Comparative effects of dietary functional nutrients on growth performance, meat quality, immune responses, and stress biomarkers in broiler chickens raised under heat stress conditions. *Anim. Biosci.* 34:1839–48
14. Lee KH. 2001. Proteomics: a technology-driven and technology-limited discovery science. *Trends Biotechnol*. 19:217–22
15. Chin L, Andersen JN, Futreal PA. 2011. Cancer genomics: from discovery science to personalized medicine. *Nat. Med.* 17:297–303
16. Barghi N, Hermisson J, Schlötterer C. 2020. Polygenic adaptation: a unifying framework to understand positive selection. *Nat. Rev. Genet.* 21:769–81

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

17. Van Goor A, Ashwell CM, Persia ME, Rothschild MF, Schmidt CJ, Lamont SJ. 2016. Quantitative trait loci identified for blood chemistry components of an advanced intercross line of chickens under heat stress. *BMC Genom*. 17:287
18. Van Goor A, Bolek KJ, Ashwell CM, Persia ME, Rothschild MF, et al. 2015. Identification of quantitative trait loci for body temperature, body weight, breast yield, and digestibility in an advanced intercross line of chickens under heat stress. *Genet. Sel. Evol.* 47:96
19. Elbeltagy AR, Bertolini F, Fleming DS, Van Goor A, Ashwell CM, et al. 2019. Natural selection footprints among African chicken breeds and village ecotypes. *Front. Genet.* 10:376
20. Fleming DS, Weigend S, Simianer H, Weigend A, Rothschild M, et al. 2017. Genomic comparison of indigenous African and Northern European chickens reveals putative mechanisms of stress tolerance related to environmental selection pressure. *G3* 7:1525–37
21. Gu J, Liang Q, Liu C, Li S. 2020. Genomic analyses reveal adaptation to hot arid and harsh environments in native chickens of China. *Front. Genet.* 11:582355
22. Xu NY, Liu ZY, Yang QM, Bian PP, Li M, Zhao X. 2022. Genomic analyses for selective signatures and genes involved in hot adaptation among indigenous chickens from different tropical climate regions. *Front. Genet.* 13:906447
23. Xie J, Tang L, Lu L, Zhang L, Xi L, et al. 2014. Differential expression of heat shock transcription factors and heat shock proteins after acute and chronic heat stress in laying chickens (*Gallus gallus*). *PLOS ONE* 9:e102204
24. Czech B, Wang Y, Wang K, Luo H, Hu L, Szyda J. 2022. Host transcriptome and microbiome interactions in Holstein cattle under heat stress condition. *Front. Microbiol.* 13:998093
25. Hu L, Sammad A, Zhang C, Brito LF, Xu Q, Wang Y. 2022. Transcriptome analyses reveal essential roles of alternative splicing regulation in heat-stressed Holstein cows. *Int. J. Mol. Sci.* 23:10664
26. Li G, Yu X, Portela Fontoura AB, Javaid A, de la Maza-Escola VS, et al. 2023. Transcriptomic regulations of heat stress response in the liver of lactating dairy cows. *BMC Genom*. 24:410
27. Li H, Zhang G, Liu Y, Gao F, Ye X, et al. 2024. Hypoxia-inducible factor 1α inhibits heat stress-induced pig intestinal epithelial cell apoptosis through eif2α/ATF4/CHOP signaling. *Sci. Total Environ.* 924:171649
28. Srikanth K, Park JE, Ji SY, Kim KH, Lee YK, et al. 2020. Genome-wide transcriptome and metabolome analyses provide novel insights and suggest a sex-specific response to heat stress in pigs. *Genes* 11:540
29. Wang Y, Wu ZW, Mou Q, Chen L, Fang T, et al. 2023. Global 3′-UTRome of porcine immature Sertoli cells altered by acute heat stress. *Theriogenology* 196:79–87
30. Jastrebski SF, Lamont SJ, Schmidt CJ. 2017. Chicken hepatic response to chronic heat stress using integrated transcriptome and metabolome analysis. *PLOS ONE* 12:30181900
31. Kim DY, Han GP, Lim C, Kim JM, Kil DY. 2023. Effect of dietary betaine supplementation on the liver transcriptome profile in broiler chickens under heat stress conditions. *Anim. Biosci.* 36:1632–46
32. Kim DY, Lim B, Kim JM, Kil DY. 2022. Integrated transcriptome analysis for the hepatic and jejunal mucosa tissues of broiler chickens raised under heat stress conditions. *J. Anim. Sci. Biotechnol.* 13:79
33. Kim H, Kim H, Seong P, Arora D, Shin D, et al. 2021. Transcriptomic response under heat stress in chickens revealed the regulation of genes and alteration of metabolism to maintain homeostasis. *Animals* 11:2241
34. Kim HS, Kim J, Kim J, Choi YH. 2022. Characterization of differential gene expression of broiler chicken to thermal stress in discrete developmental stages. *Anim. Cells Syst.* 26:62–69
35. Lan X, Hsieh JC, Schmidt CJ, Zhu Q, Lamont SJ. 2016. Liver transcriptome response to hyperthermic stress in three distinct chicken lines. *BMC Genom*. 17:955
36. Lim C, Lim B, Kil DY, Kim JM. 2022. Hepatic transcriptome profiling according to growth rate reveals acclimation in metabolic regulatory mechanisms to cyclic heat stress in broiler chickens. *Poult. Sci.* 101:102167
37. Monson MS, Van Goor AG, Ashwell CM, Persia ME, Rothschild MF, et al. 2018. Immunomodulatory effects of heat stress and lipopolysaccharide on the bursal transcriptome in two distinct chicken lines. *BMC Genom*. 19:643
38. Nan J, Yang H, Rong L, Jia Z, Yang S, Li S. 2023. Transcriptome analysis of multiple tissues reveals the potential mechanism of death under acute heat stress in chicken. *BMC Genom*. 24:459

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

39. Park W, Srikanth K, Lim D, Park M, Hur T, et al. 2019. Comparative transcriptome analysis of Ethiopian indigenous chickens from low and high altitudes under heat stress condition reveals differential immune response. *Anim. Genet.* 50:42–53
40. Pritchett EM, Van Goor A, Schneider BK, Young M, Lamont SJ, Schmidt CJ. 2023. Chicken pituitary transcriptomic responses to acute heat stress. *Mol. Biol. Rep.* 50:5233–46
41. Saelao P, Wang Y, Chanthavixay G, Yu V, Gallardo RA, et al. 2021. Distinct transcriptomic response to Newcastle disease virus infection during heat stress in chicken tracheal epithelial tissue. *Sci. Rep.* 11:7450
42. Saelao P, Wang Y, Gallardo RA, Lamont SJ, Dekkers JM, et al. 2018. Novel insights into the host immune response of chicken Harderian gland tissue during Newcastle disease virus infection and heat treatment. *BMC Vet. Res.* 14:280
43. Sun L, Lamont SJ, Cooksey AM, McCarthy F, Tudor CO, et al. 2015. Transcriptome response to heat stress in a chicken hepatocellular carcinoma cell line. *Cell Stress Chaperones* 20:939–50
44. te Pas MFW, Park W, Srikanth K, Kemp S, Kim JM, et al. 2019. Transcriptomic profiles of muscle, heart, and spleen in reaction to circadian heat stress in Ethiopian highland and lowland male chicken. *Cell Stress Chaperones* 24:175–94
45. Wang Y, Saelao P, Kern C, Jin SH, Gallardo RA, et al. 2020. Liver transcriptome responses to heat stress and Newcastle disease virus infection in genetically distinct chicken inbred lines. *Genes* 11:1067
46. Zhang Q, Lu YQ, Cao ZW, Yang YZ. 2020. Effects of heat stress on the hepatic transcriptome of yellow-feathered chickens. *S. Afr. J. Anim. Sci.* 50:626–34
47. Hua QY, Zhang MH, Wang L, Bai LL, Li L, et al. 2020. Temperature-sensitive pathways may be involved in duck embryonic developmental recovery from blastoderm dormancy during hatching. *Br. Poult. Sci.* 61:366–74
48. Kim JM, Lim KS, Byun M, Lee KT, Yang YR, et al. 2017. Identification of the acclimation genes in transcriptomic responses to heat stress of White Pekin duck. *Cell Stress Chaperones* 22:787–97
49. Yang C, Huang XB, Chen SJ, Li XJ, Fu XL, et al. 2021. The effect of heat stress on proliferation, synthesis of steroids, and gene expression of duck granulosa cells. *Anim. Sci. J.* 92:e13617
50. Reed KM, Mendoza KM, Strasburg GM, Velleman SG. 2022. Transcriptome response of proliferating muscle satellite cells to thermal challenge in commercial turkey. *Front. Physiol.* 13:970243
51. Reed KM, Mendoza KM, Xu J, Strasburg GM, Velleman SG. 2022. Transcriptome response of differentiating muscle satellite cells to thermal challenge in commercial turkey. *Genes* 13:1857
52. Ashburner M, Ball CA, Blake JA, Botstein D, Butler H, et al. 2000. Gene Ontology: tool for the unification of biology. *Nat. Genet.* 25:25–29
53. Hu C, Yang J, Qi Z, Wu H, Wang B, et al. 2022. Heat shock proteins: biological functions, pathological roles, and therapeutic opportunities. *MedComm 2020* 3:e161
54. Bar-Lavan Y, Shemesh N, Ben-Zvi A. 2016. Chaperone families and interactions in metazoa. *Essays Biochem*. 60:237–53
55. Lewis SA, Tian G, Vainberg IE, Cowan NJ. 1996. Chaperonin-mediated folding of actin and tubulin. *J. Cell Biol.* 132:1–4
56. Nakamura K, Morrison SF. 2010. A thermosensory pathway mediating heat-defense responses. *PNAS* 107:8848–53
57. Brake J, Balnave D, Dibner JJ. 1998. Optimum dietary arginine:lysine ratio for broiler chickens is altered during heat stress in association with changes in intestinal uptake and dietary sodium chloride. *Br. Poult. Sci.* 39:639–47
58. Zhu W, Jiang W, Wu LY. 2014. Dietary L-arginine supplement alleviates hepatic heat stress and improves feed conversion ratio of Pekin ducks exposed to high environmental temperature. *J. Anim. Physiol. Anim. Nutr.* 98:1124–31
59. Xia Z, Huang L, Yin P, Liu F, Liu Y, et al. 2019. L-Arginine alleviates heat stress-induced intestinal epithelial barrier damage by promoting expression of tight junction proteins via the AMPK pathway. *Mol. Biol. Rep.* 46:6435–51
60. Hu H, Dai S, Li J, Wen A, Bai X. 2020. Glutamine improves heat stress-induced oxidative damage in the broiler thigh muscle by activating the nuclear factor erythroid 2-related 2/Kelch-like ECH-associated protein 1 signaling pathway. *Poult. Sci.* 99:1454–61

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

61. Singleton KD, Wischmeyer PE. 2006. Oral glutamine enhances heat shock protein expression and improves survival following hyperthermia. *Shock* 25:295–99
62. Wu QJ, Liu N, Wu XH, Wang GY, Lin L. 2018. Glutamine alleviates heat stress-induced impairment of intestinal morphology, intestinal inflammatory response, and barrier integrity in broilers. *Poult. Sci.* 97:2675–83
63. Zhang Q, Yang Y, Lu Y, Cao Z. 2021. iTRAQ-based quantitative proteomic analyses the cycle chronic heat stress affecting liver proteome in yellow-feather chickens. *Poult. Sci.* 100:101111
64. Cohen ED, Mariol MC, Wallace RM, Weyers J, Kamberov YG, et al. 2002. DWnt4 regulates cell movement and focal adhesion kinase during *Drosophila* ovarian morphogenesis. *Dev. Cell* 2:437–48
65. Yao HH, Matzuk MM, Jorgez CJ, Menke DB, Page DC, et al. 2004. Follistatin operates downstream of Wnt4 in mammalian ovary organogenesis. *Dev. Dyn.* 230:210–15
66. Hu C, Wen J, Gong L, Chen X, Wang J, et al. 2017. Thrombospondin-1 promotes cell migration, invasion and lung metastasis of osteosarcoma through FAK dependent pathway. *Oncotarget* 8:75881–92
67. Huang WT, Chong IW, Chen HL, Li CY, Hsieh CC, et al. 2019. Pigment epithelium-derived factor inhibits lung cancer migration and invasion by upregulating exosomal thrombospondin 1. *Cancer Lett.* 442:287–98
68. Boroughs LK, Antonyak MA, Johnson JL, Cerione RA. 2011. A unique role for heat shock protein 70 and its binding partner tissue transglutaminase in cancer cell migration. *J. Biol. Chem.* 286:37094–107
69. Lang BJ, Nguyen L, Nguyen HC, Vieusseux JL, Chai RC, et al. 2012. Heat stress induces epithelial plasticity and cell migration independent of heat shock factor 1. *Cell Stress Chaperones* 17:765–78
70. Asadollahi H, Vaez Torshizi R, Ehsani A, Masoudi AA. 2022. An association of *CEP78*, *MEF2C*, *VPS13A* and *ARRDC3* genes with survivability to heat stress in an F_2 chicken population. *J. Anim. Breed Genet.* 139:574–82
71. Wolc A, Arango J, Settar P, Fulton JE, O'Sullivan NP, Dekkers JCM. 2019. Genome wide association study for heat stress induced mortality in a white egg layer line. *Poult. Sci.* 98:92–96
72. Rowland K, Ashwell CM, Persia ME, Rothschild MF, Schmidt C, Lamont SJ. 2019. Genetic analysis of production, physiological, and egg quality traits in heat-challenged commercial white egg-laying hens using 600k SNP array data. *Genet. Sel. Evol.* 51:31
73. Saelao P, Wang Y, Chanthavixay G, Gallardo RA, Wolc A, et al. 2019. Genetics and genomic regions affecting response to Newcastle disease virus infection under heat stress in layer chickens. *Genes* 10:61
74. Rowland K, Persia ME, Rothschild MF, Schmidt C, Lamont SJ. 2019. Venous blood gas and chemistry components are moderately heritable in commercial white egg-laying hens under acute or chronic heat exposure. *Poult. Sci.* 98:3426–30
75. Schaal TP, Arango J, Wolc A, Brady JV, Fulton JE, et al. 2016. Commercial Hy-Line W-36 pullet and laying hen venous blood gas and chemistry profiles utilizing the portable i-STAT® 1 analyzer. *Poult. Sci.* 95:466–71
76. Wang Y, Saelao P, Chanthavixay G, Gallardo RA, Wolc A, et al. 2024. Genomic regions and candidate genes affecting response to heat stress with Newcastle virus infection in commercial layer chicks using chicken 600K single nucleotide polymorphism array. *Int. J. Mol. Sci.* 25:2640
77. Wang T, Wei L, Wang J, Xie L, Li YY, et al. 2020. Integrating GWAS, linkage mapping and gene expression analyses reveals the genetic control of growth period traits in rapeseed (*Brassica napus* L.). *Biotechnol. Biofuels* 13:134
78. Hubbard AH, Zhang X, Jastrebski S, Singh A, Schmidt C. 2019. Understanding the liver under heat stress with statistical learning: an integrated metabolomics and transcriptomics computational approach. *BMC Genom.* 20:502
79. Li X, Zhao X, Yu M, Zhang M, Feng J. 2024. Effects of heat stress on breast muscle metabolomics and lipid metabolism related genes in growing broilers. *Animals* 14:430
80. Tu WL, Cheng CY, Chen CJ, Chan HL, Wang SH, et al. 2018. Annotation of differential protein expression in the hypothalami of layer-type Taiwan country chickens in response to acute heat stress. *J. Therm. Biol.* 77:157–72
81. Zampiga M, Laghi L, Zhu C, Cartoni Mancinelli A, Mattioli S, Sirri F. 2021. Breast muscle and plasma metabolomics profile of broiler chickens exposed to chronic heat stress conditions. *Animal* 15:100275

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

82. Tang X, Meng Q, Gao J, Zhang S, Zhang H, Zhang M. 2015. Label-free quantitative analysis of changes in broiler liver proteins under heat stress using SWATH-MS technology. *Sci. Rep.* 5:15119
83. Tu WL, Cheng CY, Chen CJ, Chan HL, Wang SH, et al. 2018. Proteomic analysis of the hypothalamus of broiler-type Taiwan country chickens in response to acute heat stress. *Anim. Sci. J.* 89:1475–85
84. Zheng HT, Zhuang ZX, Chen CJ, Liao HY, Chen HL, et al. 2021. Effects of acute heat stress on protein expression and histone modification in the adrenal gland of male layer-type country chickens. *Sci. Rep.* 11:6499
85. Ma D, Liu Q, Zhang M, Feng J, Li X, et al. 2019. iTRAQ-based quantitative proteomics analysis of the spleen reveals innate immunity and cell death pathways associated with heat stress in broilers (*Gallus gallus*). *J. Proteom.* 196:11–21
86. He J, Xia C, He Y, Pan D, Cao J, et al. 2019. Proteomic responses to oxidative damage in meat from ducks exposed to heat stress. *Food Chem.* 295:129–37
87. Zeng T, Jiang X, Li J, Wang D, Li G, et al. 2013. Comparative proteomic analysis of the hepatic response to heat stress in Muscovy and Pekin ducks: insight into thermal tolerance related to energy metabolism. *PLOS ONE* 8:e76917
88. Cheng CY, Tu WL, Chen CJ, Chan HL, Chen CF, et al. 2018. Functional genomics study of acute heat stress response in the small yellow follicles of layer-type chickens. *Sci. Rep.* 8:1320
89. Gill SP, Gangwar PC. 1990. Carbonic anhydrase and its isozymes in heat stressed white Leghorn hens. *Arch. Exp. Veterinarmed.* 44:383–88
90. Chouchani ET, Pell VR, Gaude E, Aksentijević D, Sundier SY, et al. 2014. Ischaemic accumulation of succinate controls reperfusion injury through mitochondrial ROS. *Nature* 515:431–35
91. Liu YL, Ding KN, Shen XL, Liu HX, Zhang YA, et al. 2022. Chronic heat stress promotes liver inflammation in broilers via enhancing NF-κB and NLRP3 signaling pathway. *BMC Vet. Res.* 18:289
92. Bednarczyk M, Dunislawska A, Stadnicka K, Grochowska E. 2021. Chicken embryo as a model in epigenetic research. *Poult. Sci.* 100:101164
93. Kisliouk T, Meiri N. 2009. A critical role for dynamic changes in histone H3 methylation at the Bdnf promoter during postnatal thermotolerance acquisition. *Eur. J. Neurosci.* 30:1909–22
94. Yahav S, McMurtry JP. 2001. Thermotolerance acquisition in broiler chickens by temperature conditioning early in life—the effect of timing and ambient temperature. *Poult. Sci.* 80:1662–66
95. Yahav S. 2019. Alleviating heat stress in domestic fowl: different strategies. *World's Poult. Sci. J.* 65:719–32
96. Dunislawska A, Pietrzak E, Wishna Kadawarage R, Beldowska A, Siwek M. 2022. Pre-hatching and post-hatching environmental factors related to epigenetic mechanisms in poultry. *J. Anim. Sci.* 100:skab370
97. Loyau T, Hennequet-Antier C, Coustham V, Berri C, Leduc M, et al. 2016. Thermal manipulation of the chicken embryo triggers differential gene expression in response to a later heat challenge. *BMC Genom.* 17:329
98. David SA, Vitorino Carvalho A, Gimonnet C, Brionne A, Hennequet-Antier C, et al. 2019. Thermal manipulation during embryogenesis impacts H3K4me3 and H3K27me3 histone marks in chicken hypothalamus. *Front. Genet.* 10:1207
99. Star L, Juul-Madsen HR, Decuypere E, Nieuwland MG, de Vries Reilingh G, et al. 2009. Effect of early life thermal conditioning and immune challenge on thermotolerance and humoral immune competence in adult laying hens. *Poult. Sci.* 88:2253–61
100. Fathi MM, Galal A, Radwan LM, Abou-Emera OK, Al-Homidan IH. 2022. Using major genes to mitigate the deleterious effects of heat stress in poultry: an updated review. *Poult. Sci.* 101:102157
101. Boonkum W, Chankitisakul V, Kananit S, Kenchaiwong W. 2024. Heat stress effects on the genetics of growth traits in Thai native chickens (Pradu Hang dum). *Anim. Biosci.* 37:16–27
102. Radwan LM. 2020. Genetic improvement of egg laying traits in Fayoumi chickens bred under conditions of heat stress through selection and gene expression studies. *J. Therm. Biol.* 89:102546
103. Chen ZY, Gan JK, Xiao X, Jiang LY, Zhang XQ, Luo QB. 2013. The association of SNPs in *Hsp90β* gene 5′ flanking region with thermo tolerance traits and tissue mRNA expression in two chicken breeds. *Mol. Biol. Rep.* 40:5295–306
104. Kong LN, Zhang DX, Ji CL, Zhang XQ, Luo QB. 2015. Association analysis between SNPs in the 5′-flanking region of the chicken *GRP78* gene, thermotolerance parameters, and tissue mRNA expression. *Genet. Mol. Res.* 14:6110–23

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08

105. Bettencourt BR, Hogan CC, Nimali M, Drohan BW. 2008. Inducible and constitutive heat shock gene expression responds to modification of Hsp70 copy number in *Drosophila melanogaster* but does not compensate for loss of thermotolerance in Hsp70 null flies. *BMC Biol*. 6:5
106. Shumega AR, Pavlov YI, Chirinskaite AV, Rubel AA, Inge-Vechtomov SG, Stepchenkova EI. 2024. CRISPR/Cas9 as a mutagenic factor. *Int. J. Mol. Sci.* 25:823
107. Yarnall MTN, Ioannidi EI, Schmitt-Ulms C, Krajeski RN, Lim J, et al. 2023. Drag-and-drop genome insertion of large sequences without double-strand DNA cleavage using CRISPR-directed integrases. *Nat. Biotechnol.* 41:500–12
108. Adli M. 2018. The CRISPR tool kit for genome editing and beyond. *Nat. Commun.* 9:1911
109. Whyte J, Glover JD, Woodcock M, Brzeszczynska J, Taylor L, et al. 2015. FGF, insulin, and SMAD signaling cooperate for avian primordial germ cell self-renewal. *Stem Cell Rep*. 5:1171–82
110. Ballantyne M, Woodcock M, Doddamani D, Hu T, Taylor L, et al. 2021. Direct allele introgression into pure chicken breeds using Sire Dam Surrogate (SDS) mating. *Nat. Commun.* 12:659
111. Woodcock ME, Gheyas AA, Mason AS, Nandi S, Taylor L, et al. 2019. Reviving rare chicken breeds using genetically engineered sterility in surrogate host birds. *PNAS* 116:20930–37
112. Ye X, Avendano S, Dekkers JC, Lamont SJ. 2006. Association of twelve immune-related genes with performance of three broiler lines in two different hygiene environments. *Poult. Sci.* 85:1555–69
113. Hoffmann I. 2010. Climate change and the characterization, breeding and conservation of animal genetic resources. *Anim. Genet.* 41(Suppl. 1):32–46
114. Bhawa S, Morêki JC, Machete JB. 2023. Poultry management strategies to alleviate heat stress in hot climates: a review. *J. World's Poult. Res.* 13:1–19
115. Zhou H, Baltenweck I, Dekkers J, Gallardo R, Kayang BB, et al. 2024. Feed the Future Innovation Lab for Genomics to Improve Poultry: a holistic approach to improve indigenous chicken production focusing on resilience to Newcastle disease. *World's Poult. Sci. J.* 80:273–97
116. Joost S, Colli L, Baret PV, Garcia JF, Boettcher PJ, et al. 2010. Integrating geo-referenced multiscale and multidisciplinary data for the management of biodiversity in livestock genetic resources. *Anim. Genet.* 41(Suppl. 1):47–63
117. Gheyas AA, Vallejo-Trujillo A, Kebede A, Lozano-Jaramillo M, Dessie T, et al. 2021. Integrated environmental and genomic analysis reveals the drivers of local adaptation in African indigenous chickens. *Mol. Biol. Evol.* 38:4268–85
118. Supek F, Bosnjak M, Skunca N, Smuc T. 2011. REVIGO summarizes and visualizes long lists of Gene Ontology terms. *PLOS ONE* 6:e21800

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:08